AN IN-DEPTH COMPARATIVE ANALYSIS OF SPEECH DEEPFAKE DETECTION TECHNIQUES

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February 2, 2026

ABSTRACT

The rapid progress of deep learning, combined with the widespread availability of smart audio devices, has led to an exponential increase in speech deepfake content. This term refers to a broad range of techniques that generate or manipulate audio signals, which can be exploited to commit crimes and influence public opinion by impersonating virtually everyone. In this review, we compare and contrast different techniques for detecting such fakes, examining datasets, evaluation metrics and model architectures. We then test and compare three architectures on the ASVspoof 2019 dataset: two variations of Light Convolutional Neural Network (LCNN), taking as input respectively MEL and LFCC spectrogram features, and RawFormer, a transformer-based model that processes raw audio signals directly. Experiments show stable performance across all solutions, with RawFormer outperforming LCNN variants, by achieving a 4.31% EER after just 3 training epochs. Finally, Grad-CAM explainable AI (XAI) technique is applied to the final convolutional layer of the LCNN models, showing how different spectrogram representations lead to different focus regions.

***K*eywords** Deepfake · Fake Audio Detection · Speech Synthesis · Audio Processing

1 Introduction

As of today, media content such as images, videos, audio and text can be easily generated on any device using generative AI techniques. These generated data may pose serious societal security concerns, as this content can be exploited for criminal activities, such as disinformation or identity impersonation. In a famous case of March 2019 [7] a U.K.-based energy company felt victim of a deepfake audio attack, where the attacker perfectly cloned the CEO's voice, which resulted in a loss of over $243,000 USD. Given these threats, many Speech Deepfake Detection solutions have been proposed in the literature. Early methods relied on handcrafted features using machine learning techniques, while recent solutions, with the explosion of deep learning, used techniques such as CNN, RNN, Transformers, and recently LLMs, taking as input both spectrogram representations and raw audio signals. In the meantime many datasets and benchmarks were proposed, such as the ASVspoof challenge, with many versions and different attacks proposed. In this study, we focus on reviewing the main datasets proposed for audio deepfake detection, defining the type of techniques used to create them, such as text-to-speech (TTS) and voice conversion (VC). We also define the most commonly used evaluation metrics such as Equal Error Rate and compare several architectures. Finally, we develop and compare three architectures on the ASVspoof 2019 dataset[25]: a Light Convolutional Neural Network taking as input MEL and LFCC spectrogram features, and RawFormer, a transformer-based model designed to take raw audio signals. In addition to the performance comparison, we visualize, based on XAI technique Grad-CAM, the focus regions of the last convolutional layer of both LCNN solutions, providing insight on how different spectrogram representations may influence the focus region.

The main contributions of this analysis are:

- A complete review of datasets, metrics and architectures proposed for the DSD (Deepfake Speech Detection) problem.

- An experimental comparison of spectrogram-based LCNN solutions and raw-audio transformer-based architecture on ASVspoof 2019 dataset.
- A visualization of LCNN models' focus regions based on Grad-CAM, showing differences between MEL and LFCC representations.

2 Datasets

Since 2015, major public datasets proposed for the DSD task are related to open evaluation challenges. In this section, we analyze and compare the most widely used datasets. Surprisingly, the majority of them are related to the English language, showing an important limitation and the need for broader linguistic coverage. Audio deepfake datasets are mainly generated from three techniques used to build synthetic audio: Text-to-Speech (TTS), Voice Conversion (VC) and Adversarial Attacks (AT). TTS systems generate speech directly from text, VC, on the other hand, modifies original speech to mimic the voice of a specified target speaker. Adversarial attacks, instead, try to perturb audio signals in order to deceive automatic speech processing systems. In order to compare the quality and coverage of existing datasets, we compare them based on language coverage, the number of bonafide (real) and spoofed (fake) audio samples, the types of deepfake generation techniques used and the audio sampling rate.

ASVspoof 2015[27] The ASVspoof 2015 dataset was the first edition of the ASVspoof (Automatic Speaker Verification Spoofing and Countermeasures) challenge, it was recorded in a studio environment, consisting of multiple speakers with eight different accents.

ASVspoof 2019, ASVspoof 2021[25][28] The ASVspoof 2019 and ASVspoof 2021 editions introduced 2 scenarios, LA (logical access) and PA (physical access). The LA refers to TTS and VC systems, while PA refers to replay attacks. Compared to the previous edition, they improved the realism by using audio recorded also in non-ideal and diverse conditions.

WaveFake[8] The WaveFake dataset is from 2021 and, differently from the ASVspoof series, was obtained using six different GAN-based TTS models, containing also Japanese audio samples.

CFAD[19] Moving beyond English only collections, CFAD (Chinese Fake Audio Dataset) is a publicly available Mandarin speech Deepfake dataset, generated by 12 different algorithms.

MLAAD[20] The MLAAD (Multilingual Large-scale Audio Deepfake) dataset is the best resource dataset for linguistic coverage since it is the first dataset to cover over 20 languages, making it useful for cross-linguistic detections.

ASVspoof 5[24] The ASVspoof 5 dataset is built on top of the MLS English corpus, which incorporates data from more than 4000 speakers recorded with different devices. This dataset contains a wide range of spoofing techniques and it is by far the most complete, English-focused dataset available.

Table 1: Comparison of the main public datasets for Deepfake Speech Detection (DSD)

Dataset	Year	Language(s)	# Bonafide	# Spoof	Techniques	Sampling Rate
ASVspoof 2015 [27]	2015	EN	16651	246500	TTS | VC	16 kHz
ASVspoof 2019 (LA) [25]	2019	EN	12483	108978	TTS | VC	16 kHz
ASVspoof 2021 (LA) [28]	2021	EN	18452	163114	TTS | VC	Multiple
WaveFake [8]	2021	EN | JAP	0	117985	TTS	16 kHz
MLAAD [20]	2024	23	0	76000	TTS	22 kHz
CFAD [19]	2024	CH	0	374000	TTS | VC	16 kHz
ASVspoof 5 [24]	2024	EN	188819	815262	TTS, VC, AT	8/16 kHz

3 Metrics

In this section we compare metrics specifically used in speech Deepfake detection literature. It is important to notice that other metrics, such as Precision, Recall, Accuracy, F1-score are used as well.

EER The EER (Equal Error Rate) is the principal metric used for Spoofing Countermeasure (CM), it explicitly tells how well the detector is good at separating real audio from fake audio. It is the point where the False Acceptance Rate (FAR) is equal to the False Rejection Rate (FRR). FAR measures the percentage of fake audio that are marked as real, while FRR indicates the percentage of real speech marked wrongly as fake. The lower EER is, the better.

t-DCF[14] The t-DCF (Tandem Detection Cost Function) is a metric introduced in ASVspoof 2019, it evaluates a combined system, an ASV (Automatic Speaker Verification) system protected by a CM (Spoofing Countermeasure). The system scheme can be conceptualized as:

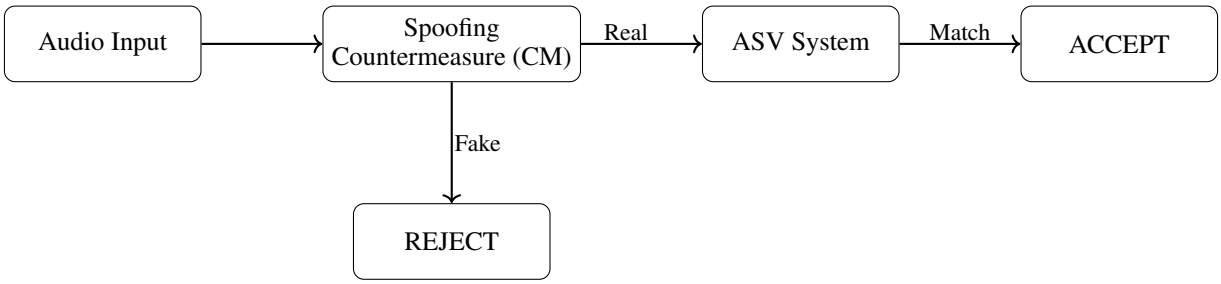


Figure 1: Tandem architecture for t-DCF metric

It measures the reliability of the ASV system when it is controlled by the spoofing protection layer. The lower t-DCF is, the better, 1 means the system is useless, 0 means perfect. The formula of the metric is computed as:

$$\text{t-DCF}(t_{\text{cm}}, t_{\text{asv}}) := C_{\text{miss}}\, \pi_{\text{tar}}\, P_{\text{miss,tar}}(t_{\text{cm}}, t_{\text{asv}}) + C_{\text{fa,non}}\, \pi_{\text{non}}\, P_{\text{fa,non}}(t_{\text{cm}}, t_{\text{asv}}) + C_{\text{fa,df}}\, \pi_{\text{df}}\, P_{\text{fa,df}}(t_{\text{cm}}, t_{\text{asv}}) \tag{1}$$

Equation (1) defines the following parameters:

Class Priors (π): π_{tar}, π_{non}, and π_{df} are the priors for the ASV targets, ASV non-targets, and deepfake attacks, which sum up to 1.

Costs (C): C_{miss}, $C_{\text{fa,non}}$, and $C_{\text{fa,df}}$ are the penalties cost for rejecting a target, accepting a non-target, and accepting a deepfake, respectively.

Error Rates (P): $P_{\text{miss,tar}}$, $P_{\text{fa,non}}$, and $P_{\text{fa,df}}$ represent the detection error rates.

a-DCF[12] The a-DCF (Architecture-agnostic Detection Cost Function) is a variant of t-DCF more flexible, it requires a single threshold t instead of two, arguing that joint optimization of ASV and CM components can enhance spoofing robustness, as the strength of one subsystem can compensate for the other.

$$\text{a-DCF}(t_{\text{cm}}, t_{\text{asv}}) := C_{\text{miss}} \cdot \pi_{\text{tar}} \cdot P_{\text{miss}}(t) + C_{\text{fa,non}} \cdot \pi_{\text{non}} \cdot P_{\text{fa,non}}(t) + C_{\text{fa,df}} \cdot \pi_{\text{df}} \cdot P_{\text{fa,df}}(t) \tag{2}$$

Equation (2) defines the following parameters:

Class Priors (π): π_{tar}, π_{non}, and π_{df} are the priors for the ASV targets, ASV non-targets, and deepfake attacks, which sum up to 1.

Costs (C): C_{miss}, $C_{\text{fa,non}}$, and $C_{\text{fa,df}}$ are the penalties cost for rejecting a target, accepting a non-target, and accepting a deepfake, respectively.

Error Rates (P): P_{miss}, $P_{\text{fa,non}}$, and $P_{\text{fa,df}}$ represent the false alarm rates for non-target speaker verification and Deepfake spoofs, respectively. The a-DCF has been used as the primary metric for the ASVspoof5 challenge [24].

4 Architectures overview

Deepfake Speech Detection architectures have evolved over the recent years. First solutions were focusing on traditional machine learning algorithms, such as Support Vector Machines (SVM), Gaussian-based models and fusion techniques such as Gaussian Mixture Models (GMMs)[11], typically operating on handcrafted features such as Mel-Frequency Cepstral Coefficients (MFCC). With the diffusion of deep learning, newer data-driven models emerged. Convolutional neural networks (CNN) were the first architectures used to explore spatial features from time-frequency inputs. Many successful alternatives include ResNet variations as SE-net [15], light CNN (LCNN) [16], and RawNet [26]. To capture

temporal-features, several solutions were proposed using Recurrent Neural Networks, such as LSTMs [3]. Interesting solutions were also proposed using Graph Neural Networks (GNNs), where frequency bins and time frames were used as nodes to form a fully connected graph. More recently their variant such as Graph Convolutional Network (GCN) [1] and Graph Attention Networks (GAT) [23] were tested successfully, GCN divided usually the spectrogram in patches and extracts embedding using a CNN, GAT instead introduced the attention mechanism during node feature aggregation. Finally many solutions assessed the ability of encoder-decoder transformers [4], and its combination with other architectures such as CNN and ResNet.

Table 2: Public architectures performances on 2019 ASVspoof LA dataset

Year	Architecture name	Loss	# Feature	EER (%)
2019	LCNN fusion [16]	A-Softmax	CQT/LFCC/DCT	1.84
2021	CNN-ResNet-MLP [10]	CE	-	1.64
2021	GAT [23]	CE	LFB-spec	1.68
2022	1-D Conv. Transformer [17]	Focal Loss	LFCC-spec	1.06
2023	GCN [1]	CE	LFB-spec	0.58
2023	RawFormer [18]	CE	-	0.59

4.1 Feature Extraction vs End-to-End Systems

DSD systems are divided in two broad categories based on what type of input data the architecture takes. The first category refers to end-to-end (E2E) systems that takes directly raw audio 1D tensor in input, while the second category refers to systems that takes hand-crafted features as input, called feature-based or two-stage systems. Feature-based systems transforms firstly the 1D audio tensor into a 2D (time-frequency) tensor, using like the Short Time Fourier Transform (STFT)[21]. On top of that 2D tensor, a first group of features consists of spectrogram representation, like the linear-frequency STFT spectrogram or the MEL spectrogram. The STFT spectrogram maintains the linear frequency resolution and does not apply any filtering. On the contrary, MEL spectrogram applies a Mel-scale filter bank on the STFT magnitude, emphasizing certain frequencies considered more important. A second group of features includes cepstral representations, such as Mel-Frequency Cepstral Coefficients (MFCC) [5] and Linear Frequency Cepstral Coefficients (LFCC) [13]. These features are computed using a filter bank (a Mel-scale for MFCC and a linear-scale for LFCC) to the STFT spectrum, followed by a log compression and a Discrete Cosine Transform (DCT). The DCT decorrelates the filter-bank energies, resulting in a low-dimensional, very rich representation, which showed to be very effective for DSD tasks. Recently, feature extraction has been performed using pre-trained self-supervised model such as HuBERT [9] and WavLM [2], which learn rich speech representation automatically from unlabeled data. These models can be used directly frozen or fine-tuned for our specific task. One of the major limitation of feature-based approach, with respect of using directly raw waveforms, is the loss of information in the extraction of the features, which can lead in poorer results, for this reason many recent solutions involves E2E systems that takes as input the raw audio directly, letting the model to learn its useful patterns. The downside of this greater flexibility is the probability of having higher computational complexity models.

5 Experiments

In order to dive deep in the analysis of detection architectures, we build and analyze three representative solutions, each reflecting a different design idea in terms of input representation or network composition:

- **LCNN with Mel Spectrogram features**[16]: a light CNN, which takes as input MEL extracted features.
- **LCNN with LFCC features**[13] a light CNN, which takes as input the Linear Frequency Cepstral Coefficients (LFCC) extracted features.
- **RawFormer**[18]: an end-to-end model which models positional-related local and global dependencies in raw audio wave-forms.

All the experiments are conducted on the ASVSpoof2019 dataset [25], in particular on the LA (Logical Access) subset, excluding the PA (physical access) subset. LA refers to scenarios where the audio is directly fed into the system, while PA refers to scenarios where audios are replayed through speakers (Replay Attacks).

5.1 LCNN solutions

LCNN, which stands for Light Convolutional Neural Networks is a prominent architecture tested for the task of audio deepfake classification. LCNN differs from standard CNN for the fact that the activation (e.g. ReLU) is removed and substituted by the so-called Max-Feature-Map layer. We test in particular MFM 2/1 operation which combines two feature maps and outputs their element-wise maximum, selecting the dominant feature for each location.

$$y_{ij}^{k} = \max\left(x_{ij}^{k}, x_{ij}^{k+N}\right) \tag{3}$$

where $x \in \mathbb{R}^{2N \times H \times W}$ per-sample input yields $y \in \mathbb{R}^{N \times H \times W}$, for $1 \leq k \leq N$, $1 \leq i \leq H$, $1 \leq j \leq W$.

5.1.1 MEL spectrogram features

In our first experiment, we train a LCNN taking as input a time-frequency representation using the MEL spectrogram representation. The Mel spectrogram applies a non-linear transformation on top of the STFT power spectrum in order to approximate the human auditory system. The STFT spectrum stands for Short-Time Fourier Transform, it window the 1-D raw audio signal into overlapping short segments and apply the fourier transform on each one. The Mel transformation is achieved by projecting the magnitude spectrum onto a Mel-scale filterbank, consisting of various overlapping triangular filters. The relationship between the frequency f (in Hertz) and the perceptual pitch m (in Mels) is defined, based on the HTK (Hidden Markov Toolkit) formulation as:

$$m = 2595 \cdot \log_{10}\left(1 + \frac{f}{700}\right) \tag{4}$$

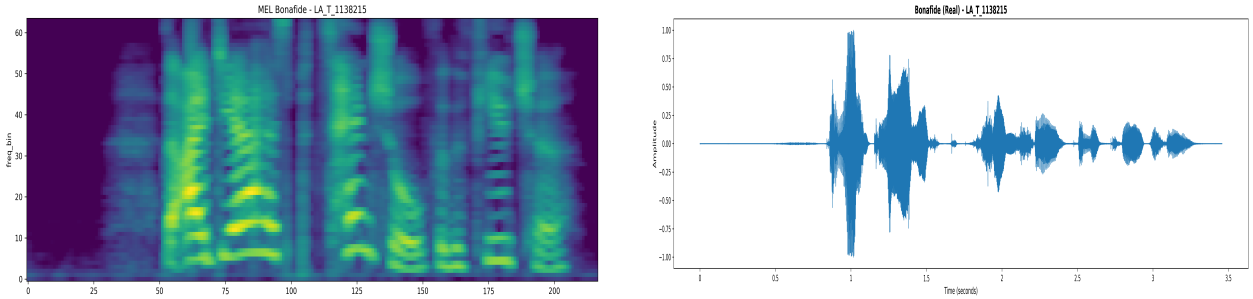
By applying this filterbank, the high-dimensional linear frequency axis is compressed into a lower-dimensional representation that preserves perceptually relevant spectral structures while discarding high-frequency fine details.

Sample Rate $= 16,000Hz:$ the sampling frequency of the input raw audio.

$N_{mels} = 64:$ the number of Mel filters. This results in an output tensor of 64.

Target Frames $= 600$**:** fixed temporal resolution for consistent LCNN input.

STFT Parameters: $n_{\text{fft}} = 512$, window length 320, hop length (160).



(a) Mel spectrogram

(b) Raw audio signal

Figure 2: Comparison Mel spectrogram - 1D audio signal

5.1.2 LFCC spectrogram features

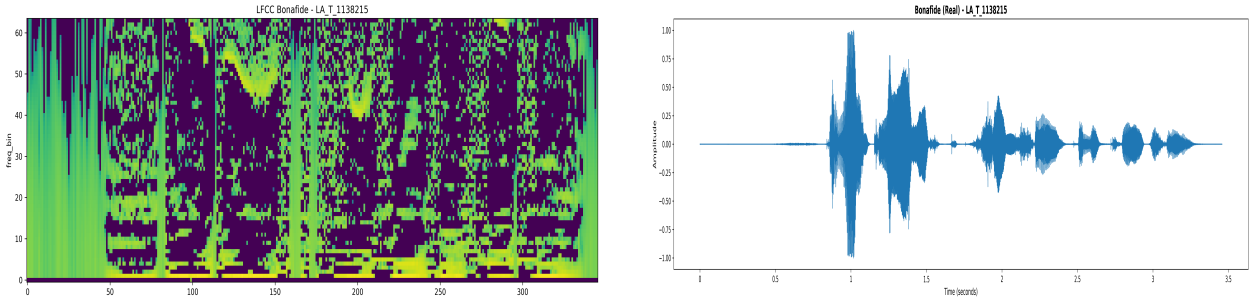
In our second experiment, we train a LCNN taking as input LFCC features. The LFCC representation applies a linear-frequency filterbank to the STFT magnitude spectrum, capturing smooth spectral details. Unlike Mel features which use triangular filter, LFCC computes linear-frequency cepstral coefficients via a discrete cosine transform of the log-power spectrum. The transformation pipeline follows: STFT magnitude → log power spectrum → linear filterbank (without Mel warping) → DCT to obtain the cepstral coefficients.

Sample Rate $= 16,000\ Hz$**:** the sampling frequency of the input raw audio.

$N_{\textbf{lfcc}} = 64$**:** the number of LFCC coefficients. This results in an output tensor of 64.

Target Frames $= 600$**:** fixed temporal resolution for consistent LCNN input.

STFT Parameters: $n_{\text{fft}} = 512$, window length 320, hop length 160.



(a) LFCC spectrogram

(b) Raw audio signal

Figure 3: Comparison LFCC spectrogram - 1D audio signal

5.1.3 LCNN architecture

The networks for both Mel and LFCC inputs are equal, since the dimensionality of input is the same. The models alternate convolutions, MFM activation, max-pooling and batch norm shrinking the time-frequency resolution. After the convolutional backbone, features go through a fully connected layer, followed by a dropout and batch norm. A final fully connected layer output bonafide vs spoofed binary prediction. As showed in 3, the MFM blocks include a 3x3 convolution with padding=1 and stride=1, followed by a max-feature map operation to halve the channels.

Table 3: Architecture of the LCNN systems. K=Kernel, S=stride, P=padding

Layer	**K / S / P**	**Output Shape**	**Param**
Input	-	$1 \times 64 \times 600$	-
Conv2d	5×5 / 1 / 2	$64 \times 64 \times 600$	1.6K
MFM	3×3 / 1 / 1	$32 \times 64 \times 600$	36.9K
MaxPool	2×2 / 2	$32 \times 32 \times 300$	0
Conv2d	1×1 / 1	$64 \times 32 \times 300$	2.1K
MFM	3×3 / 1 / 1	$32 \times 32 \times 300$	36.9K
BN	-	$32 \times 32 \times 300$	64
Conv2d	3×3 / 1 / 1	$96 \times 32 \times 300$	27.7K
MFM	3×3 / 1 / 1	$48 \times 32 \times 300$	83.0K
MaxPool	2×2 / 2	$48 \times 16 \times 150$	0
BN	-	$48 \times 16 \times 150$	96
Conv2d	1×1 / 1	$96 \times 16 \times 150$	4.7K
MFM	3×3 / 1 / 1	$48 \times 16 \times 150$	83.0K
BN	-	$48 \times 16 \times 150$	96
Conv2d	3×3 / 1 / 1	$128 \times 16 \times 150$	55.4K
MFM	3×3 / 1 / 1	$64 \times 16 \times 150$	147.6K
MaxPool	2×2 / 2	$64 \times 8 \times 75$	0
BN	-	$64 \times 8 \times 75$	128
Conv2d	1×1 / 1	$128 \times 8 \times 75$	8.3K
MFM	3×3 / 1 / 1	$64 \times 8 \times 75$	147.6K
BN	-	$64 \times 8 \times 75$	128
Conv2d	3×3 / 1 / 1	$64 \times 8 \times 75$	36.9K
MFM	3×3 / 1 / 1	$32 \times 8 \times 75$	36.9K
MaxPool	2×2 / 2	$32 \times 4 \times 37$	0
BN	-	$32 \times 4 \times 37$	64
Flatten	-	4736	-
FC1	-	160	0.76M
MFM	-	80	0
FC2	-	2	162
Total			**1.47M**

5.2 RawFormer

In our third experiment, we train the RawFormer [18], which takes as input raw audio waveforms directly. In particular, we tested the RawFormer-S variant, composed of four two-dimensional convolutional blocks and two Transformers encoder layers, used to capture the local-global dependency for anti-spoofing. In particular, a RawNet2 [22] architecture is used as front-end, with a Sinc-based convolutional layer, which extracts specific frequency bands from raw waveforms using pre-defined filters and produces higher-level feature maps (HFM) for learning local dependencies, then, via a 1D Positional Aggregator and Transformer-based classifier global dependencies are captured too.

5.2.1 1D Positional Aggregator

The higher-level feature maps produced by the front-end are flattened along the time and frequency axes. To encode positional information, a one-dimensional sinusoidal positional encoding is applied to the resulting sequence. The positional encoding is defined as:

$$P(x, 2i) = \sin\left(\frac{x}{1000^{2i/C}}\right), \quad P(x, 2i+1) = \cos\left(\frac{x}{1000^{2i/C}}\right),$$

where x is the index of the combined frequency, i the channel index and C the channels. Subsequently this block with positional information is put inside the transformers encoders composed of multi-head attention and feed forward modules. Finally embeddings are sent into a sequence pool layer and a final linear layer which produces the output.

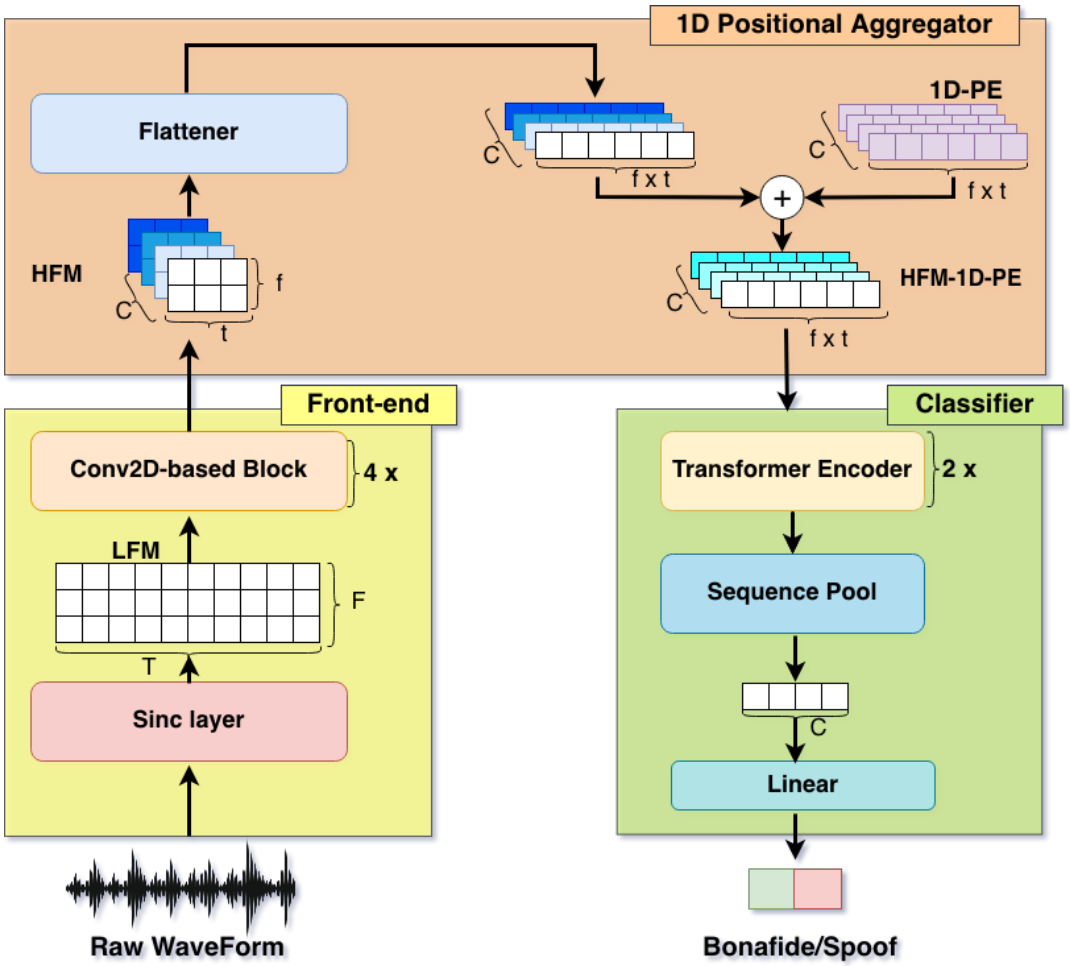


Figure 4: Architecture of the RawFormer-S model.
C, f, and t denote the number of channels, frequency bins, and time frames, respectively.

6 Performance comparison

All the solutions were trained on a Kaggle P100 GPU for 3 epochs on the full training set, consisting of 25,380 audio files. The evaluation is performed on the entire evaluation set, consisting of 71,237 audio files.

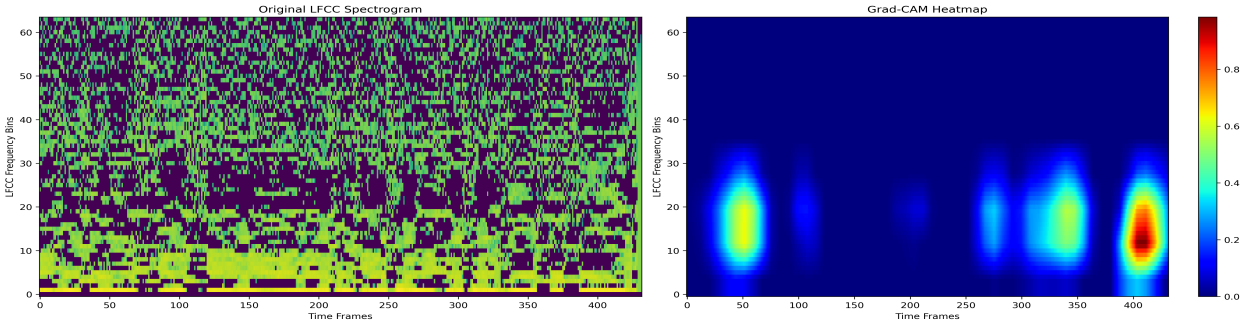
Table 4: Performance comparison of the models

MODEL	EER (%)	Accuracy (%)	Precision (%)	Recall (%)	F1 (%)
LCNN-MEL	6.85	94.20	**98.83**	94.66	**96.60**
LCNN-LFCC	11.16	87.04	44.17	**99.60**	60.10
RawFormer-S	**4.31**	**96.26**	75.4	94.6	84.10

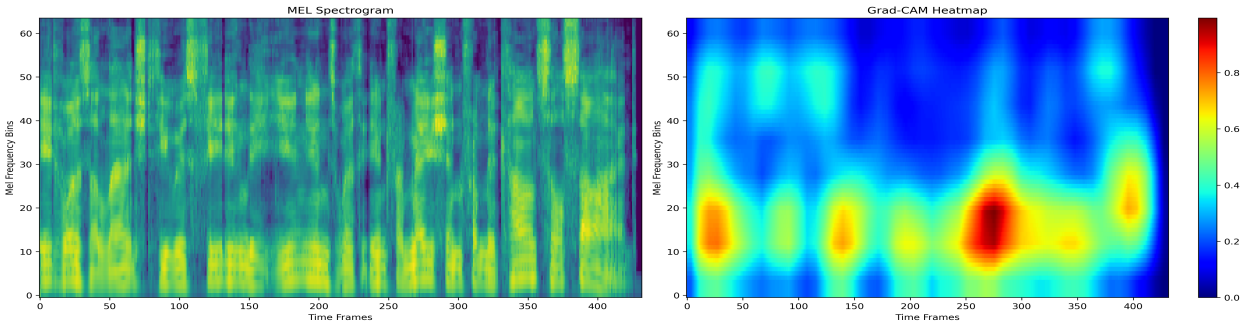
As shown in Table 4, RawFormer-S achieves the best performance among the three models, obtaining the lowest EER of 4.31% and the highest accuracy of 96.26%, showing the strongest capability of distinguishing between spoofed and bonafide samples. The LCNN-MEL model obtains the highest precision and F1 score, but its slightly higher EER of 6.85% suggests weaker overall class separability. The LCNN-LFCC model performs the worst, obtaining the highest EER. Looking at the recall (99.6%), it shows a very good capability to detect spoofed samples, but at the cost of many false positives. Overall, none of the models reaches near state-of-the-art performance when trained for only 3 epochs, but the results show how different architectures can change detection capabilities and the intrinsic limits of CNN only architectures in catching spoofing details.

7 Grad-CAM Visualizations

For the final experiment, we used Grad-CAM (Gradient-weighted Class Activation Map) to generate heatmaps of the most relevant regions in the last convolutional layer of the LCNN networks, to evaluate whether different spectrograms lead to different focus regions. Grad-CAM creates the heatmaps by weighting the final convolutional layer's feature maps with gradients of the target class score; then it upsamples the activation map via interpolation to match the input spectrogram. As shown in 5, 6, 7, LCNN MEL and LFCC networks focus on similar regions, but not exactly the same, matching our intuition that different spectrograms may hide or show important details for detecting spoofed audio.

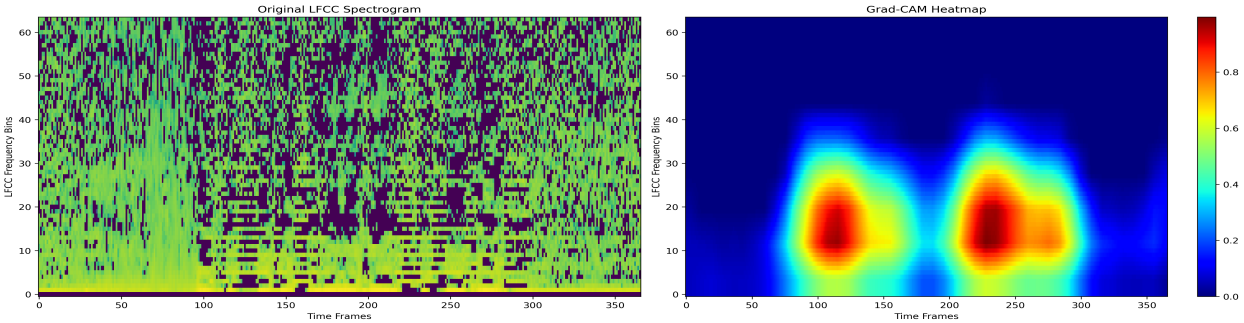


(a) LFCC spoof sample - Spectrogram and Grad-CAM heatmaps

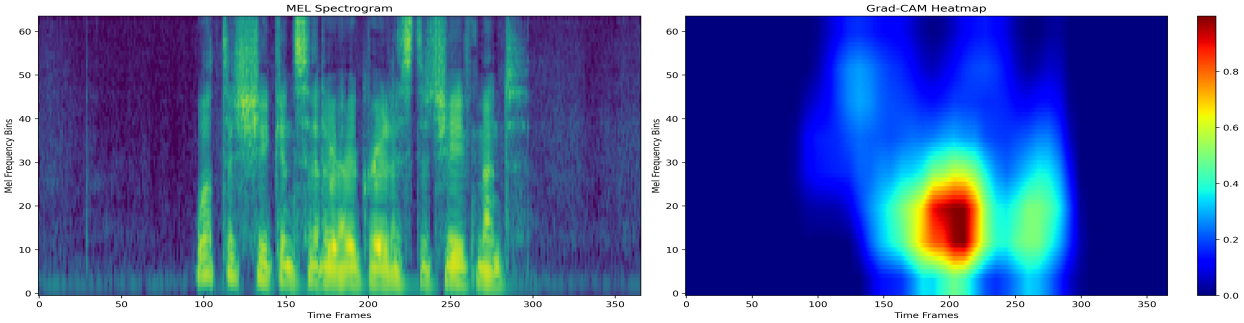


(b) Mel spoof sample - Spectrogram and Grad-CAM heatmaps

Figure 5: Heatmap comparison of MEL-LFCC features - first example

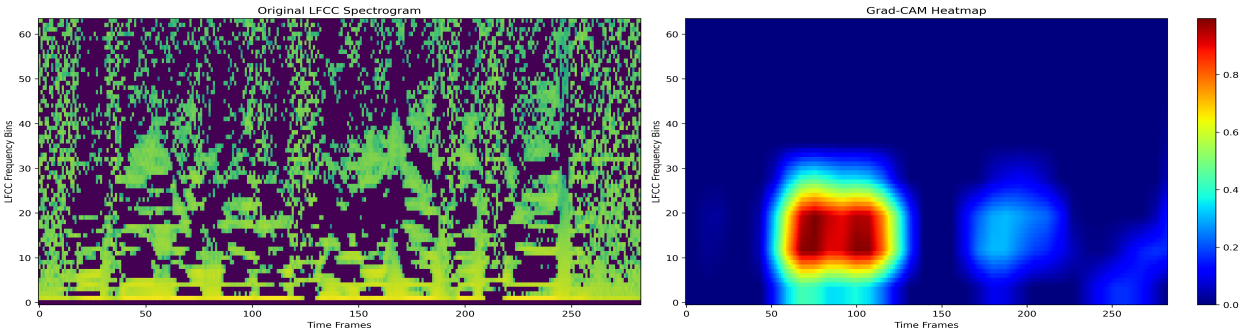


(a) LFCC spoof sample - Spectrogram and Grad-CAM heatmaps

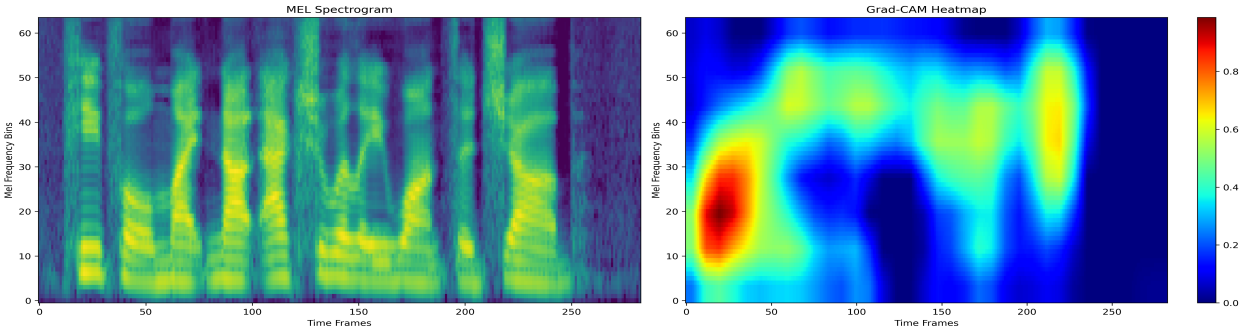


(b) Mel spoof sample - Spectrogram and Grad-CAM heatmaps

Figure 6: Heatmap Grad-CAM comparison of MEL-LFCC features - second example



(a) LFCC spoof sample - Spectrogram and Grad-CAM heatmaps



(b) Mel spoof sample - Spectrogram and Grad-CAM heatmaps

Figure 7: Heatmap Grad-CAM comparison of MEL-LFCC features - third example

8 Limitations and Ethical Aspects

8.1 Limitations

In recent years deepfake audio detectors have made notable progresses, however several key limitations persist:

- Most performing detection models usually rely on large-scale deep learning architectures, making them not suitable for mobile and embedded devices. Future work could address this limitation through model pruning techniques or other solutions for real-time detection.
- The majority of available datasets are still limited to the English language, lacking sufficient coverage of real-world scenarios and highlighting the need for broader multilingual datasets.
- Many state-of-the-art models operate as a black box, lacking interpretability. This makes it difficult to identify spoofing techniques and to detect false positives.
- Even if modern architectures obtain near perfect results on in-distribution data, out-of-distribution data, obtained using different techniques or by using different domains, still make those models struggle, evidencing the need for more generalization.

8.2 Ethical Aspects

With the rapid expansion of deepfake generated audios, many ethical challenges are posed, including the risk of misinformation and disinformation, erosion of trust in audiovisual material, and harm to individual and companies through impersonation and fraud. Deepfake audio/image content is explicitly targeted in the EU AI act, in the article 50 [6]. In particular, Deepfakes are considered as Limited-Risk Systems, mandating that any AI-generated or altered media must be clearly disclosed (e.g. via labels, watermarks or metadata) and technically detectable. These safeguards become compulsory by August 2026, with bans on abusive manipulation enforced immediately and expensive fines up to 35M € or 7% of the global turnover. Detection models introduce further ethical concerns. Explainability is critical as black-box models like deep neural networks often lack transparency, making it very hard to justify the decisions in high risk scenarios such as legal evidence or elections. Imbalance training data can create fairness and bias problems, as most trainable datasets are in English and detection models may perform poorly on diverse accents or dialects.

9 Conclusion

This survey provides a broad overview of the deepfake audio detection landscape. It reviews the main datasets, metrics, and architectures, highlighting their strengths and remaining limitations. In addition, we evaluate three architectures to show how the choice of input representations (raw waveforms and spectrograms) and architectural components can lead to big jumps in performances. In our experiments, RawFormer, which operates directly on raw waveforms, achieved the best results compared to the two LCNN variants based on spectrogram features. Despite the strong performances of many state-of-the-art models, many challenges remain not solved as cross-data generalization, adversarial robustness and real-time detection. As generative audio improves, with the risk of near-perfect fakes in the next few years, there is an urgent need for adaptive, explainable systems and new benchmarks to push forward the research community.

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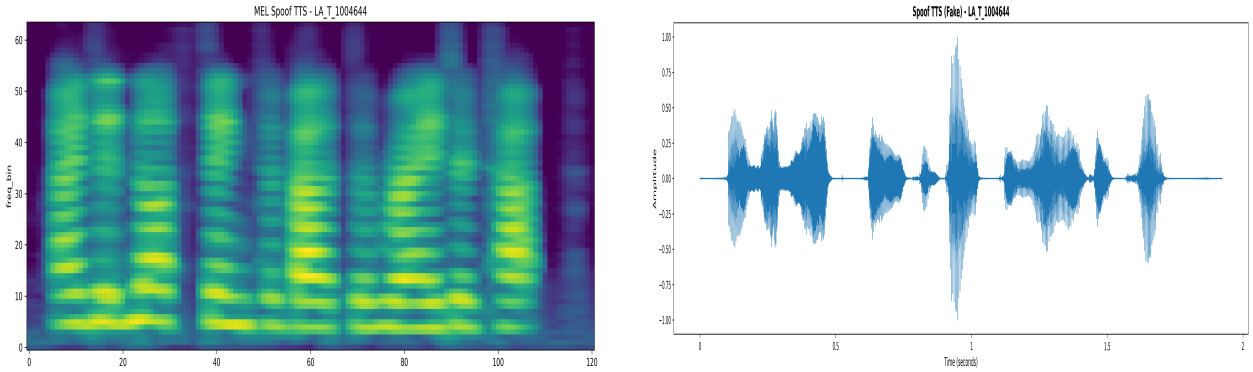
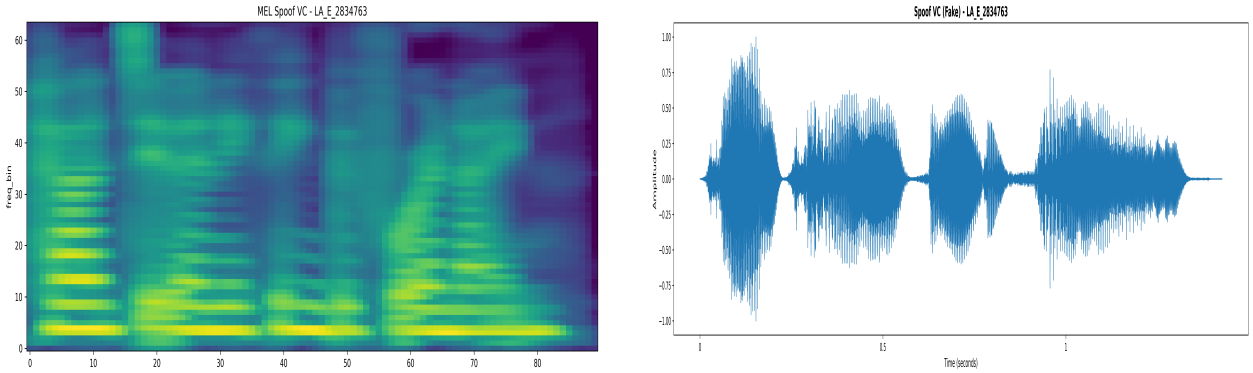
A Spectrogram comparison - TTS and VC Spoofed Audio

A.1 MEL Spectrogram comparison TTS-VC



(a) Mel spectrogram (b) Raw audio signal

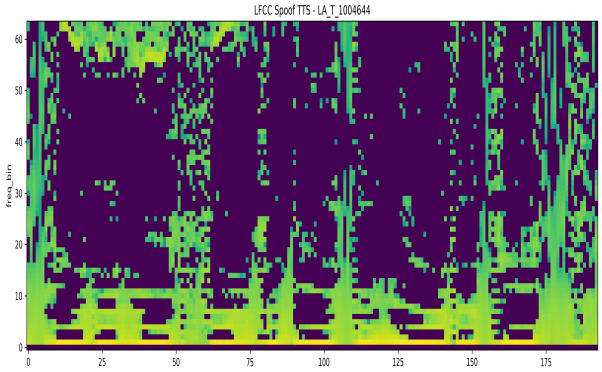
Figure 8: Comparison Mel spectrogram - 1D audio signal TTS



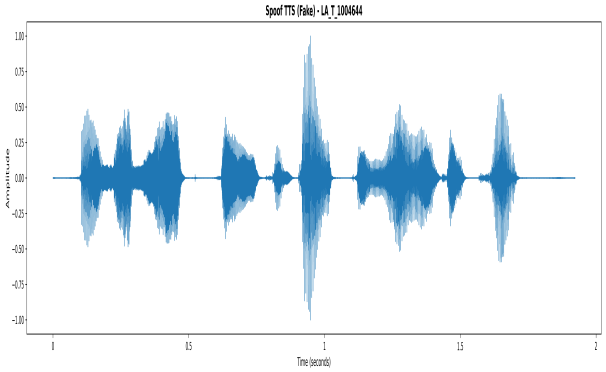
(a) Mel spectrogram (b) Raw audio signal

Figure 9: Comparison Mel spectrogram - 1D audio signal VC

A.2 LFCC Spectrogram comparison TTS-VC

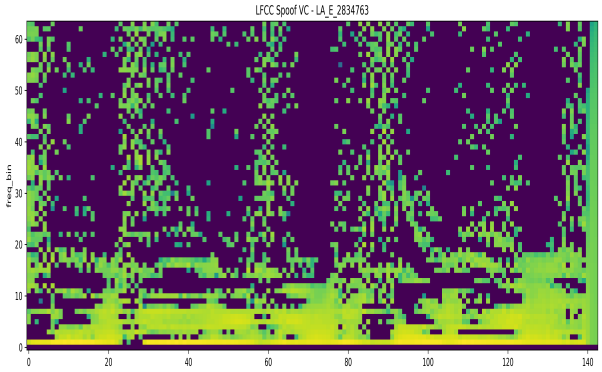


(a) LFCC spectrogram

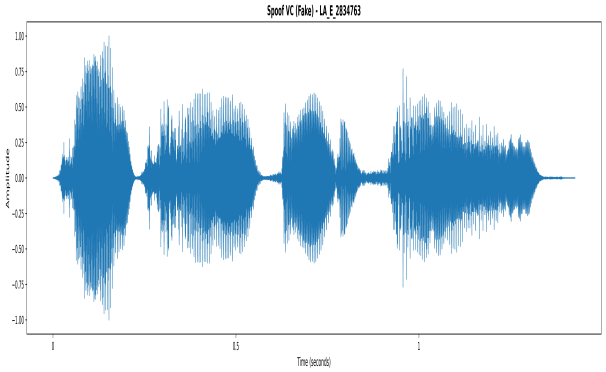


(b) Raw audio signal

Figure 10: Comparison LFCC spectrogram - 1D audio signal TTS



(a) LFCC spectrogram



(b) Raw audio signal

Figure 11: Comparison LFCC spectrogram - 1D audio signal VC